

Task 'Unit 8 Refactoring and Code Smells' Unit 8

Task Collaborative Discussion 4

The code snippet includes two main code smells, magic numbers and rigid conditional logic, the discount values (0.9 and 0.8) are also hardcoded, and this reduces readability because their meaning is unclear, they also reduce maintainability, because changing discounts requires manual edit of the function. This issue is resolved by replacing magic numbers with named constants. The additional issue is the if-elif-else structure, even though the method is short, the conditional chain violates the Open/Closed Principle because every new item type requires the existing logic to be modified. This reduces scalability and increases maintenance effort, a suitable refactoring technique is the strategy pattern, which separates each discount rule into its own class. strategy pattern improves maintainability in systems where pricing rules are expected to expand.

(Lacerda, et al, 2020).

```
```python
```

```
BOOK_DISCOUNT = 0.90
```

```
ELECTRONICS_DISCOUNT = 0.80
```

```
class NoDiscount:
```

```
 def apply(self, price):
```

```
 return price
```

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```
class BookDiscount:
```

```
 def apply(self, price):
```

```
 return price * BOOK_DISCOUNT
```

```
class ElectronicsDiscount:
```

```
 def apply(self, price):
```

```
 return price * ELECTRONICS_DISCOUNT
```

```
DISCOUNTS = {
```

```
 'book': BookDiscount(),
```

```
 'electronics': ElectronicsDiscount()
```

```
}
```

```
def calculate_total_price(items):
```

```
 total = 0
```

```
 for item in items:
```

```
 strategy = DISCOUNTS.get(item['type'], NoDiscount())
```

```
 total += strategy.apply(item['price'])
```

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```
return total
```

```
'''
```

This version improves readability, removes hardcoded values and eliminates conditional business logic from the main function. New discount types can be added without changing existing code, improving extensibility and maintainability.

(Lacerda, et al, 2020)

### References

- Lacerda, G., Petrillo, F., Pimenta, M. and Guéhéneuc, Y.G., 2020. Code smells and refactoring: A tertiary systematic review of challenges and observations. *Journal of Systems and Software*, 167, p.110610.

### Use of Digital Tools

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## **Task 'Unit 8 Refactoring and Code Smells' Unit 8**

work of the author. AI tools were used for exploration, same way as academic literature exploration. Any use of content mentioned in this document is properly referenced to its original author and was used responsibly to ensure integrity and originality. I have used Artificial Intelligence responsibly and ethically, in accordance with the assignment brief. All AI use has been declared and evidenced.